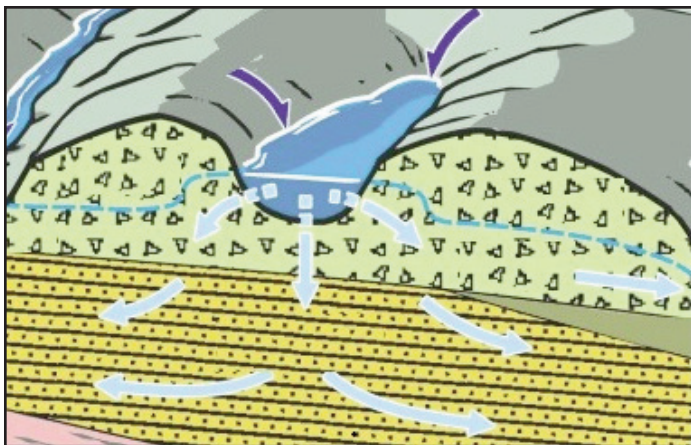


REPORT



Groundwater Science and Policy in Canada: Takeaways from an International Association of Hydrogeologists (IAH) Canada Conference Panel in 2025

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SUMMARY

Groundwater provides potable water for approximately a third of all Canadians, sustains two-thirds of the annual baseflow of many Canadian rivers and underpins agricultural, industrial and ecosystem functions in every province and territory. Yet

groundwater has received disproportionately little attention in federal freshwater science, monitoring and policy compared to surface water. The low visibility of groundwater as a topic complicates groundwater governance and hinders advancements in resource assessment, scientific understanding and national reporting of data and research, especially given the already complex division of federal, provincial and territorial responsibilities related to water resources.

In spring 2025, the Geological Survey of Canada, the Canada Water Agency and the International Association of Hydrogeologists – Canadian National Chapter (now IAH Canada) co-organized a panel discussion on *Public groundwater science and policy in Canada: Challenges of jurisdiction, funding and capacity*. The panel was held at the annual Geological Association of Canada – Mineralogical Association of Canada meeting in Ottawa, which was jointly sponsored by IAH Canada. The panel brought together federal, provincial and territorial government representatives, conservation practitioners and academic hydrogeologists to address four themes: 1) overcoming political and policy barriers; 2) funding and capacity; 3) data, science and monitoring needs; and 4) public awareness.

This article summarizes that panel session, providing context through a brief review of the past 55 years of federal groundwater policy conversations. It highlights issues raised by the panellists related to governance barriers, monitoring, funding, capacity and communication, and concludes that four key outcomes need to be pursued: 1) the inclusion of groundwater in any renewed federal freshwater policy framework, 2) stable, multi-year funding for groundwater science and monitoring, 3) development of a national monitoring network, 4) formalization of groundwater expertise within federal freshwater decision-making processes.

RÉSUMÉ

Les eaux souterraines fournissent de l'eau potable à environ un tiers de la population canadienne, soutiennent les deux tiers de l'écoulement de base annuel de nombreuses rivières canadiennes et sont à la base des fonctions agricoles, industrielles et écosystémiques dans chaque province et territoire. Pourtant, les eaux souterraines ont reçu une attention disproportionnellement faible dans la science, la surveillance et les politiques fédérales relatives à l'eau douce, comparativement aux eaux de surface. La faible visibilité des eaux souterraines en tant que sujet complique leur gouvernance et freine les avancées en matière d'évaluation des ressources, de compréhension scientifique et de rapport national des données et de la recherche —

d'autant plus que le partage des responsabilités fédérales, provinciales et territoriales à l'égard des ressources en eau est déjà complexe.

Au printemps 2025, la Commission géologique du Canada, l'Agence canadienne de l'eau et l'Association internationale des hydrogéologues – Chapitre national canadien (aujourd'hui IAH Canada) ont coorganisé une table ronde intitulée « Science et politique publiques des eaux souterraines au Canada : enjeux de compétence, de financement et de capacité ». Cette table ronde s'est tenue lors du congrès annuel conjoint de l'Association géologique du Canada et de l'Association minéralogique du Canada, à Ottawa, événement également commandité par IAH Canada. Elle a réuni des représentants des gouvernements fédéral, provinciaux et territoriaux, des praticiens de la conservation et des hydrogéologues universitaires autour de quatre thèmes : 1) surmonter les obstacles politiques et réglementaires; 2) le financement et la capacité; 3) les besoins en matière de données, de science et de surveillance; et 4) la sensibilisation du public.

Cet article résume cette séance, en offrant une mise en contexte par un bref survol des 55 dernières années de discussions sur les politiques fédérales relatives aux eaux souterraines. Il met en évidence les enjeux soulevés par les panélistes concernant les obstacles de gouvernance, la surveillance, le financement, la capacité et la communication, et conclut que quatre résultats clés doivent être visés : 1) l'inclusion des eaux souterraines dans tout cadre renouvelé de politique fédérale sur l'eau douce; 2) un financement stable et pluriannuel pour la science et la surveillance des eaux souterraines; 3) l'élaboration d'un réseau national de surveillance; 4) la formalisation de l'expertise en matière d'eaux souterraines au sein des processus décisionnels fédéraux relatifs à l'eau douce.

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INTRODUCTION

Groundwater is an indispensable source of freshwater and a critical component of the water cycle (Fig. 1). It plays a key role in both natural systems and socio-economic development. About a third of all Canadians depend on groundwater sources for potable water, including two-thirds of those living in rural areas (Council of Canadian Academies 2009). Other sources suggest the figure is closer to 30% (Kohut et al. 2014). The reliance exceeds 90% in Yukon and reaches 100% in Prince Edward Island. More detailed information on water use by source, sector and region is provided by Agrawal et al. (2022).

In Canada, groundwater is typically closely linked to stream flow, bodies of water and aquatic ecosystem health. During low-flow seasons, groundwater contributes most of stream flow (known as 'base flow'). Globally, groundwater contributes more to agricultural irrigation than surface water when considering the contribution of groundwater to surface water flows (Wood and Cherry 2021).

Despite the importance of groundwater, it remains a largely overlooked component of the water cycle, partly because it exists out of sight, making it hard to quantify and understand (e.g. Nowlan 2005; United Nations 2022), and partly because it

is complex to manage, as it is not necessarily confined by watershed boundaries, political jurisdictions or divisions of responsibility.

Canada is renowned for its abundant supplies of freshwater; nevertheless, the country's federal governance structure and legislative assignments of water responsibility create a complex setting for water action. Canada's challenges around water management are far from unique (e.g. Megdal et al. 2015; Dudley Ward 2024; Madani 2026). The United States, European Union and other federations face similar difficulties managing jurisdictional responsibilities for water resources (e.g. Gerlak 2005; Boscheck 2006; Petersen-Perlman et al. 2018).

Over the years, many observers have asserted the need for more effective water governance and management in Canada. Several have specifically noted the federal government's potential to harmonize legislation and reduce fragmentation (Bakker 2007; Hill et al. 2008). As a step toward practical change, in 2019 the federal government initiated action to create the Canada Water Agency.

A panel at the 2025 joint meeting of the Geological Association of Canada, Mineralogical Association of Canada, and International Association of Hydrogeologists – Canadian National Chapter (GAC-MAC-IAH-CNC) in Ottawa reflected on this context to discuss public groundwater science and policy in Canada, focusing on challenges of jurisdiction, funding and capacity.

This article provides highlights from the last 55 years of Canadian federal freshwater policy, includes a brief chronology of the development of the Canada Water Agency, summarizes key points made during the panel session, and identifies persistent issues and concerns that affect groundwater governance in Canada, drawing on both the panel session and the broader policy record. It goes on to propose four specific ways to substantively address those issues and concerns, noting where all the above align with the recommendations of the May 2026 report from the Standing Committee on Environment and Sustainable Development, *Sustaining Canada's Freshwater for Today and Tomorrow* (House of Commons of Canada 2026).

A more detailed report on the panel proceedings, including full bibliographic information and panelist presentations, will be made available as a Geological Survey of Canada publication (Russell et al. in prep).

WATER IN CANADA

General Information

Canada is fortunate to have a comparative abundance of water, including 20% of the world's total freshwater reserve and 7% of the annual global renewable water supply (Environment and Climate Change Canada 2020). Groundwater is an important component of that abundance and is often indivisible from surface water (Freeze and Cherry 1979; Winter et al. 1998; Hinton 2014), contributing up to two-thirds of the annual flow of many Canadian rivers (Sharpe et al. 2014). Almost all of Canada's (and the world's) unfrozen freshwater is groundwater.

Hydrological cycle emphasizing groundwater - surface water interactions

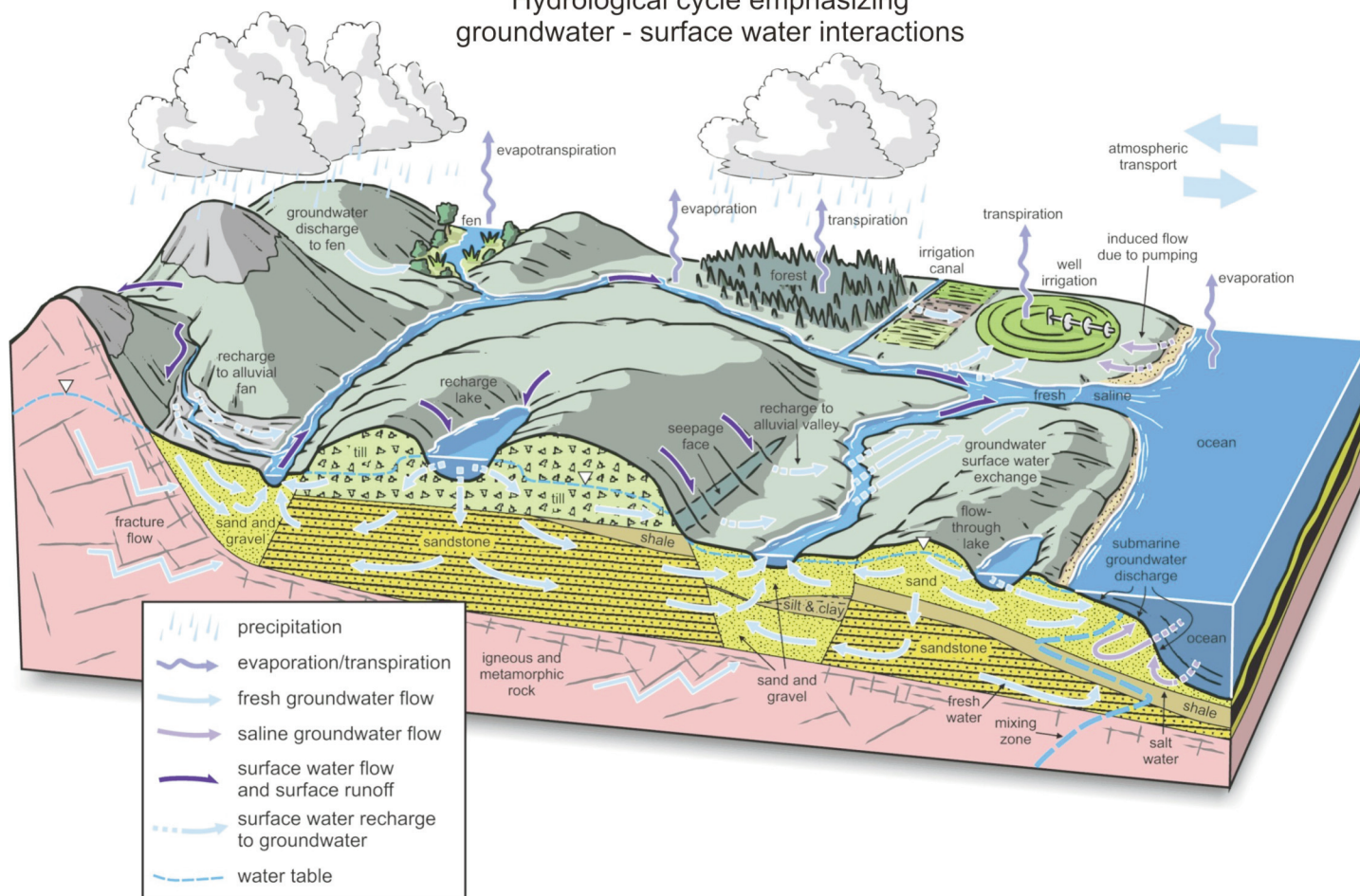


Figure 1. Water cycle highlighting groundwater–surface water components, from Hinton (2014).

Groundwater maintains minimum summer and winter flows (Paznekas and Hayashi 2016) that support aquatic ecosystem functions and broader watershed dynamics (e.g. Hinton 2014; Linnansaari et al. 2023). This ecological role stems from the continuous contribution of groundwater to surface streams in the form of discharge, and from groundwater's relatively constant temperature throughout the year, which mirrors the mean annual air temperature.

The subsurface connectivity of groundwater through pores and fractures combined with its slow velocity also means it can transport contaminants to discharge points such as rivers and lakes years after initial release, potentially for extended periods of time and over large areas. Conversely, surface water may contribute to groundwater quantity and quality by recharging aquifers in some regions (Fig. 2). Contaminated groundwater can have serious negative impacts on the health of humans, animals and associated ecosystems either directly by affecting the quality of water supply wells or indirectly by degrading surface water quality (e.g. Van Stempvoort et al. 2022; Storb et al. 2023; Hodgins et al. 2025).

Throughout much of the world, population growth, agricultural demand and industrial water use are stressing water systems and threatening water supplies (Madani 2026). Cana-

da, despite its 'water wealth', is experiencing the consequences of climate change through cyclical droughts (e.g. Cook et al. 2007) and some of the fastest rising average temperatures in the world. Canada's rate of temperature increase above pre-industrial values is roughly twice the global average (e.g. Zhang et al. 2019; <https://www.canada.ca/en/environment-climate-change/services/environmental-indicators/temperature-change.html>).

The socio-economic importance of groundwater, its ecological significance and intensifying water stress make groundwater management critically important (e.g. Council of Canadian Academies 2009; Trainer et al. 2010). Yet today, Canada has no comprehensive national system for groundwater oversight or monitoring (Ryan et al. 2026).

Freshwater Policy in Canada

Canada's provincial and territorial governments have primary jurisdiction over natural resources, including most water management and protection within their boundaries (e.g. Bakker and Cook 2011). This complicates the federal role in water governance and policy. The federal government has responsibilities related to protected areas, fish and fish habitat protection, species at risk and aquatic invasive species (*Canada Water*



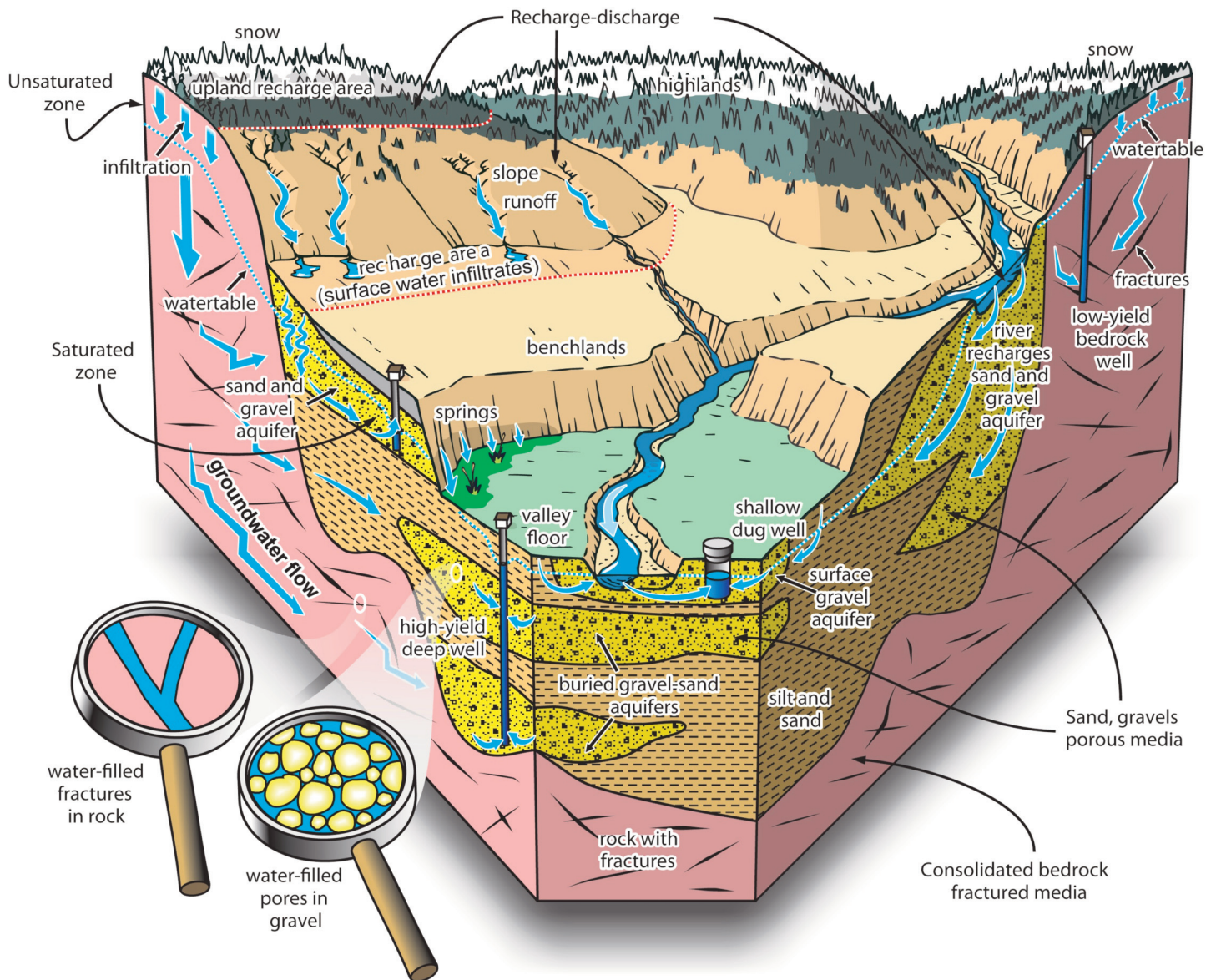


Figure 2. Illustration of groundwater flow and geological units forming aquifers. (From Rivera 2014).

Act 1985; Coté 2004). It is also responsible for navigation and shipping and waters on federal lands, and shares responsibilities with the provinces and territories for monitoring surface water, monitoring and preventing pollution, and managing inland fisheries. It is also involved in decisions when a body of water spans or flows through more than one jurisdiction, whether domestically or internationally.

The *Canada Water Act* of 1970 (hereafter, “the act”) is often seen as a centrepiece of federal water legislation, although practically it is part of a tapestry of various related laws and has been under strain since its inception (e.g. Pearse et al. 1985; Hania 2024). The overarching objective of the act is to provide for the management of water resources, including research, planning, and implementation of programs for conservation, development and utilization. The act has four distinct parts:

Part 1: Comprehensive Water Resource Management

Part 2: Water Quality Management

Part 3: Regulating Nutrients

Part 4: General Administration of the Act (covering inspectors and analysts, advisory committees, offences and punishments, parliamentary reporting requirements and more).

Part 2 has never been used. Part 3 was repealed in 1985 and is now part of the *Canadian Environmental Protection Act*, 1999 (Environment and Climate Change Canada 2025).

The diversity of interests, roles and participants in water issues at the federal level is evident in the fact that more than 20 federal departments and agencies have freshwater-related responsibilities, further complicating the legislative context (Canada Water Agency 2025a). Well over a dozen federal acts have freshwater implications, including the *Canada Water Act*, the *Fisheries Act*, the *Canadian Environmental Protection Act* and

the *Canadian Navigable Waters Act*. Provinces and territories have their own freshwater laws and associated bodies, ranging in function from primary legislation to source protection frameworks and strategic policies (e.g. Hill et al. 2008).

The first major inquiry into federal water policy under the *Canada Water Act* was initiated in the 1980s when the Minister of the Environment appointed an advisory committee that became the *Inquiry on Federal Water Policy*. A concluding report with recommendations was authored by Pearse et al. (1985).

The subsequent *Federal Freshwater Policy* (Environment Canada 1987) set out federal policy objectives for water along with a plan for assessments. Two assessments were completed (1990, 1994), only one of which is readily accessible (Interdepartmental Committee on Water 1990). Explicit recommendations were made related to groundwater issues including data management, monitoring, quality protection guidelines and integrated management. Specifically, the 1990 review recommended:

“...consultation with other departments, [and] co-ordinated strategies for the sustainable development of ground-water resources and surface resources dependent on ground water.”

Further reviews of the *Federal Freshwater Policy* were never completed, and the Interdepartmental Committee on Water became inactive around 1995 (Pearse and Quinn 1996). Some of this decline in reporting and interdepartmental action may be attributable to the 1994–1996 federal government Program Review, which resulted in significant staff and funding cuts (e.g. Tellier 2022).

Groundwater and the *Canada Water Act*

Environment and Climate Change Canada is the lead department for the *Canada Water Act* and produces an annual *Canada Water Act* report. (The department was formerly named Environment Canada and is indicated as such where appropriate throughout this article.) Coverage of groundwater varies considerably across the 55 years of those reports, possibly reflecting broader governmental priorities and focus as well as the impact of funding pressures. Historical cycles of support and funding for activities under the *Canada Water Act* were documented in the mid-1990s by Booth and Quinn (1995); by the time of their review, which fell on the 25th anniversary of the act, dedicated funding had diminished to a fraction of the original allocation.

A review of *Canada Water Act* annual reports (e.g. Environment Canada 1986; Environment and Climate Change Canada 2025) shows that groundwater work has typically focused on ecological function, anthropogenic contaminants and groundwater–surface water interactions, although a broader range of topics such as geothermal energy and groundwater supply have been considered at times. The reports tend to focus on Environment Canada / Environment and Climate Change Canada groundwater projects rather than on broader federal activities that align with and support Part 1 of the act, such as Agriculture and Agri-Food Canada’s *Drought Monitor* or the groundwater-related work of the Geological Survey of Canada

and the Canada Centre for Remote Sensing. Collaborative initiatives between Environment and Climate Change Canada and Natural Resources Canada are not discussed, nor are 30 years of pre-2023 groundwater work at Natural Resources Canada, despite joint publications (e.g. Rivera et al. 2003) and multiple consecutive interdepartmental memoranda of understanding.

Natural Resources Canada’s freshwater science mandate is set out in the *Department of Natural Resources Act* (1994) and the *Resources and Technical Surveys Act* (1985). The Geological Survey of Canada, which is part of Natural Resources Canada, is involved specifically with subsurface natural resources and as such is the centre of expertise for groundwater mapping and characterization, similar to geological survey organizations in other G7 countries.

As noted in the 2026 report from the House of Commons Standing Committee on Environment and Sustainable Development (House of Commons of Canada 2026), Environment and Climate Change Canada’s groundwater focus generally relates to water *quality* whereas Natural Resources Canada’s focus relates to groundwater *quantity*. However, these two aspects are closely interconnected, as geochemistry helps to provide a better understanding of groundwater flow, and water supply issues rely not only on quantity but also quality.

Coordination between Environment and Climate Change Canada and Natural Resources Canada seems to have been challenging. Despite a call to action by the Interdepartmental Committee on Water in 1990, 30 years of subsequent *Canada Water Act* annual reports show limited evidence of coordinated interdepartmental strategies between the two federal departments. This lack of coordination was further highlighted in a report from a panel chaired by Dr. John Cherry (Morgan 1993). That said, groundwater research has been conducted collaboratively by individual scientists from both organizations.

Growing Calls for Action

Federal freshwater research and policy in Canada have historically tended to focus on the most accessible, easiest-to-study components of the water cycle such as atmospheric water and surface water. Other parts of the water cycle have received less attention. This is at least partly because the *Canada Water Act* treats the water cycle holistically, without distinguishing between groundwater, surface water or other components such as precipitation, snow cover, glaciers or soil moisture.

The result has been a systemic imbalance in funding allocations, reporting attention, science studies and infrastructure development. For example, there are nearly 3,000 freshwater stream flow monitoring stations in Canada’s collaborative federal/provincial National Hydrometric Network (NHN) but only nine groundwater sites (Environment and Climate Change Canada 2025). Currently, government groundwater monitoring programs are conducted at the provincial, territorial or municipal levels, resulting in significant variations in monitoring station density, data types, technologies and practices for reporting results. This makes it challenging to assess groundwater quantity and quality changes at the federal level.

New issues have emerged since the last update to the *Canada Water Act* in 1985 and are not reflected in it, particularly with respect to First Nations water governance rights (Marshall et al. 2018; Centre for Indigenous Environmental Resources 2024). Academics (de Loë 2008) and non-governmental organizations (e.g. Nowlan 2005; Morris et al. 2007) have published commentaries on various aspects of federal water governance since at least the early 2000s, raising concerns about groundwater pollution, water quantity, climate impacts, First Nations water challenges and other factors.

To address the growing need for coherent, coordinated freshwater management, the Government of Canada committed in 2019 to establish a Canada Water Agency, issuing instructions through a Ministerial Mandate letter to Environment and Climate Change Canada (Office of the Prime Minister 2019).

In March 2021, the House of Commons Standing Committee on Environment and Sustainable Development (ENVI) initiated a study of federal policies and legislation relating to freshwater (House of Commons of Canada 2021). Across 22 sessions in a process that extended into 2024, government representatives, academics, private sector organizations and non-governmental organizations made 179 presentations and submitted 112 briefs (House of Commons of Canada 2021, 2024). Of those briefs, 42 reference groundwater, including, most notably, the submission from International Association of Hydrogeologists Canada (IAH Canada 2024), which recommended inclusive terminology, a national monitoring network and reporting as well as educational initiatives to raise awareness and promote a better understanding of groundwater.

Following a hiatus due to the 2025 federal election, the final report on the committee's hearings was published in May 2026 (House of Commons of Canada 2026). It recognizes the role of groundwater in the Canadian waterscape and includes a dedicated section on the topic. It also provides three recommendations regarding groundwater (recommendations 17, 18 and 19), namely to:

- dedicate sufficient attention and resources to groundwater's role in water security and climate adaptation.
- recognize the strategic value of groundwater and work with other levels of government and scientists to protect it from depletion and pollution.
- ensure groundwater is included explicitly in a modernized *Canada Water Act*.

An additional recommendation (recommendation 55) references groundwater in the context of human health and data collection but is less pertinent to the present article.

EMERGENCE OF THE CANADA WATER AGENCY

The federal government's announced intention to develop a Canada Water Agency in 2019 was followed by an extensive consultative process that engaged more than 2,700 Canadians. The findings of those consultations were published in a *What We Heard* report, which made explicit reference to groundwater, including: a) the observation that “*improved understanding and*

protection of groundwater is important to Canadians...”; b) acknowledgement of the importance of groundwater to agriculture; and c) recognition of the need for more hydrologic monitoring, including of groundwater (Environment and Climate Change Canada 2021). All provinces and territories participated, along with representatives from non-governmental organizations (NGOs), First Nations, Inuit and Métis communities.

The Canada Water Agency became a standalone entity officially in October 2024, with a mandate to improve freshwater management in Canada by providing leadership and effective collaboration federally, and by improving coordination and collaboration with provinces, territories, and Indigenous peoples to proactively address national and regional transboundary freshwater challenges and opportunities.

Table 1 summarizes key milestones in the development of the Canada Water Agency from the issuance of the original Ministerial Mandate letter to the enactment of the *Canada Water Agency Act* (2024) and ongoing freshwater strategy work. Clear from this sequence of events is the agency's foundational commitment to engagement and consultation.

THE 2025 GROUNDWATER PANEL

Organization and Design

In discussions with the Geological Survey of Canada, the Canada Water Agency suggested holding a panel on groundwater issues at the joint GAC-MAC-IAH-CNC meeting in spring 2025. This was seen as an opportunity to further highlight the importance of groundwater, build consensus among key players and acknowledge some of the challenges associated with groundwater monitoring, reporting and education, many of which have roots in Canada's complex freshwater policy environment.

The panel discussion was part of a day-long series of technical sessions on groundwater that also included presentations on groundwater monitoring, open access groundwater data, demographic and professional trends in hydrogeology, provincial groundwater geoscience, case studies from the provincial Oak Ridges Moraine Groundwater Program and groundwater studies at the Geological Survey of Canada, Natural Resources Canada (Bingham-Koslowski 2025).

The panel was moderated by Cynthia McClain, a University of Calgary hydrogeology adjunct professor and IAH Alberta Director. Panellists included Gemma Boag (Director General, Canada Water Agency), Angela Coleman (General Manager, Conservation Ontario), Brendan Mulligan (Senior Scientist, Government of Yukon), Alfonso Rivera (Chair, IAH Transboundary Aquifers Commission), and Sonia Talwar (Director General, Geological Survey of Canada).

The two-hour session included a brief round of introductions, responses to prepared questions, an open Q&A with members of the audience, and a short wrap-up provided by IAH member and University of Calgary hydrogeology professor Cathy Ryan. The prepared questions for the panel were organized in four thematic segments (Table 2). The remainder of this section summarizes key points of discussion related to each principal theme.

Table 1. Progression of the development of the Canada Water Agency and associated publications since 2019.

Year	Event
2019	Ministerial Mandate Letter directs the Minister of Environment and Climate Change Canada to create the Canada Water Agency (Office of the Prime Minister 2019)
2020	PlaceSpeak public engagement platform launched to host online consultations
2020	Government of Canada launches consultations on new Canada Water Agency
2020	Regional virtual consultation sessions held
2020	Toward the Creation of a Canada Water Agency discussion paper (Environment and Climate Change Canada 2020)
2021	Formal public and stakeholder comment period closed
2021–2023	Indigenous engagement process held with First Nations, Métis and Inuit communities
2021	Toward the Creation of a Canada Water Agency Stakeholder and public engagement – What we heard (Environment and Climate Change Canada 2021)
2021	Throne Speech and Budget reaffirm the federal government's commitment to create the Canada Water Agency
2023	Canada Water Agency established as a branch within Environment and Climate Change Canada
2023	<i>Canada Water Agency Act</i> tabled as part of Bill C-59
2024	Synthesis of Freshwater Science in Canada (Environment and Climate Change Canada 2024)
2024	National Freshwater Data Strategy workshop held at the Canada Centre for Inland Waters in Burlington, Ontario
2024	Bill C-59 receives Royal Assent on June 20
2024	Canada Water Agency launched as a standalone entity
2024–2025	About the <i>Canada Water Act</i> Review — the <i>Canada Water Act</i> (1985) renewal process — begins with pre-engagement of provinces, territories and Indigenous partners
2025	<i>National Freshwater Data Strategy</i> discussion paper based on the 2024 workshop (Canada Water Agency 2025c)
2026	<i>National Freshwater Science Agenda</i> (Environment and Climate Change Canada 2026)
2026	<i>A Force of Nature: Canada's Strategy to Protect Nature</i> announcing the Canada Water Agency will collaborate to develop Canada's first National Water Security Strategy

Panel members agreed that bringing groundwater into the freshwater conversation is important and must include not only specialists but also policymakers, industry decision makers and members of the public. That kind of engagement requires building up broad 'water literacy': widespread understanding of freshwater and the specific role groundwater plays in human health, ecosystem health and many sectors of the economy.

First theme: Overcoming political and policy barriers to groundwater management

Different groups of interest holders tend to have unique concerns and information needs about groundwater. Given this, engagements need to be tailored to their interests—for example, focusing on questions about water supplies, water quality and water resource management when working with local officials and community groups that tend to be concerned with practical community outcomes. Emphasizing the connections between scientific research and such pragmatic objectives is likely to be highly valuable and effective in such situations.

The National Dialogue on Groundwater was identified as an effective coordination mechanism for public servants working in groundwater. Created and led by the Geological Survey of Canada over the past 10 years, its goal is to improve communication among the federal, provincial and territorial governments. Panellists indicated the National Dialogue on Groundwater is Canada's only current national-scale groundwater coordination effort, although the Canada Water Agency is working to strengthen federal coordination and collaboration through mechanisms such as the Federal Freshwater Committee (which involves 20-plus federal departments and agencies with freshwater-related responsibilities) and the Federal Freshwater Hub (Canada Water Agency 2025a). The Committee brings together senior management; whereas the Hub embeds staff from other federal departments or agencies with water-related responsibilities into the Canada Water Agency. The panellists offered some thoughts on what a coordinating body or structure should ideally be able to do, observing that the National Dialogue on Groundwater is limited to facilitating information-sharing among the provinces, territories and fed-



Table 2. Themes and questions addressed by the panel. Questions have been paraphrased with complete questions available in Russell et al. (in preparation).

First theme: Overcoming political and policy barriers to groundwater management

- Question 1. What steps are needed to elevate the importance of groundwater among government decision makers to support policy action and sustainable water management?
- Question 2. What coordination mechanisms already exist? What gaps or barriers remain to bridging different water disciplines and organizational silos to support better coordination of groundwater policy, monitoring, characterization and management across Canada?
- Question 3. Is there good communication on groundwater between the three orders of government (federal, provincial/territorial and municipal)? What steps can be taken to remove barriers and improve communication?
-

Second theme: Funding and capacity challenges in groundwater

- Question 1. Why is it difficult to obtain long-term funding for groundwater monitoring, science and management, and what changes are needed to improve funding stability? What specific mechanisms (e.g. federal funding streams, provincial initiatives, public-private partnerships) could help secure stable, long-term funding for groundwater monitoring, science, and management in Canada?
- Question 2. Where is groundwater capacity needed (e.g. leadership, policy, science, monitoring), and how can this be achieved?
-

Third theme: Data, science and monitoring needs

- Question 1. Is there a need for a national groundwater monitoring network in Canada, and what are recommended approaches? Are there other key challenges (e.g. equity, jurisdiction, policy, etc.) to a national-scale monitoring approach?
-

Fourth theme: Public awareness and future directions

- Question 1. How can we increase public awareness of groundwater and its critical role in sustaining surface water, droughts, permafrost, drinking water resources and ensuring long-term water security, including for transboundary aquifers?
- Question 2. Please tell a short, impactful story about a groundwater issue.
-

eral government and does not undertake or coordinate joint projects or initiatives.

In considering alternative models, panellists discussed the National Administrators Table of the National Hydrometric Program and the Ontario Source Water Protection Framework. Panel members saw the National Administrators Table's defined mandate, budget and bilateral agreements (pertaining to hydrometrics) as key strengths. Ideally, groundwater could be folded into the National Administrators Table to improve coordination at the national level. The Ontario Source Water Protection Framework was also considered notable for its multi-stakeholder, multidisciplinary nature. Local committees of policy and technical experts work together to implement solutions, providing a structure that allows for dialogue and information exchanges that otherwise might not occur. In such a model, success hinges on ensuring that committee work is meaningful and has real influence on implementation, and that technical outputs are subject to peer review. Supporting this kind of model requires 'infrastructure' in the sense of knowledge, resources, relationships and broad representation of sectors and disciplines.

On the question of communication, panellists acknowledged challenges resulting from the complexities of many water-related issues as well as the complicated policy and gov-

ernance environment. Individual institutions have different norms and expectations around communication and different bodies of knowledge to draw on. Broad, formal engagement efforts are essential, and communication among parties must also be formalized, which will support not only knowledge sharing but also, ultimately, the implementation and management of groundwater initiatives.

Second theme: Funding and capacity challenges in groundwater

Panellists generally felt that groundwater remains a 'blind spot' for many funders, noting its exclusion from many calls for proposals on freshwater. Resources are often directed to surface water research because any changes in surface water or human health impacts related to it are visible and therefore attract political and public interest. If groundwater were more widely recognized as a significant source of freshwater and an important contributor to surface water—making it integral to overall water security and economic growth—groundwater science would draw greater institutional support and, as a result, increased funding. Issues such as conjunctive water and land use management and transboundary waters offer strong 'business cases' of the sort that could conceivably capture greater political will.



One area of activity on which the panellists agreed was the need for stable, long-term funding for groundwater monitoring. Monitoring enables scientists and decision makers to understand, protect and sustainably manage freshwater resources. Sustainable, long-term monitoring requires ongoing dialogue among federal and provincial/territorial governments so that federal decision makers can better understand regional and local needs and provincial/territorial authorities can be more aware of when funding is available. Current federal funding models for groundwater science need to change, as the existing grant structure supports single, one-off projects, not long-term, nationwide initiatives such as the operation of monitoring networks.

To the question of where capacity is needed, the panel identified at least two areas: 1) monitoring and management of water resources that are not subject to regulations (e.g. ecological flows); and 2) increased capacity at the leadership level of government. The implication of the second point is that higher level managers should have a better understanding of groundwater to make informed decisions. Canada is strong in water science but less so in groundwater policy and management. Groundwater scientists need to be integrated into freshwater decision-making processes from end to end to ensure that groundwater is properly considered.

Third theme: Data, science and monitoring needs

Given the funding-related discussion about monitoring outlined above, panel members were unified in the view that Canada needs a national groundwater monitoring network to address changing systems. The National Hydrometric Service was considered to be a good example of what is possible: its network monitors surface water, primarily water quantity. Due to jurisdictional responsibilities and funding limitations, groundwater likely requires a 'network of networks' rather than a single national network. The existing Groundwater Information Network (https://gin.geosciences.ca/service/api_ngwds:gin2/en/gin.html) could provide the foundation for this, though work remains to be done to harmonize and standardize data collection, analysis and reporting. It may be ideal to integrate groundwater monitoring into the National Administrators Table and National Hydrometric Program.

Although freshwater *management* is a provincial/territorial responsibility, the panel made the distinction that monitoring is not necessarily a management function, but rather a scientific necessity to support management. The panel suggested that the federal government should see a role for itself in groundwater monitoring as an activity to be co-led with provincial and territorial partners (as it does for surface water).

Fourth theme: Public awareness and future directions

Groundwater is challenging for many people to think about or conceptualize because they do not experience it directly. Even those who depend on it (for example, groundwater wells for household use) tend not to have a complete understanding of it. Informing people about the economic, social and environmental value of freshwater, including groundwater, is critical for informed and responsible decision-making. The Canada

Water Agency has specifically made this kind of education part of its workstreams.

At the time of the panel discussion, the 25th anniversary of the Walkerton tragedy in Ontario was approaching (O'Connor 2002). That high-profile case of groundwater contamination led to seven deaths and more than 2 000 illnesses due to the presence of groundwater-sourced *E. coli* bacteria in the town's drinking water system. Walkerton was mentioned as a story that remains a powerful testament to the importance of good groundwater science and management. Another example raised was the 2024 Eagle Gold Mine disaster in Yukon. A landslide of crushed rock treated with processing solutions caused cyanide to leach into the environment, and the subsequent contamination of water resources occurred, as it often does, through groundwater.

Audience Q&A

Following the prepared questions, panellists took questions from the audience. These ranged from challenging assumptions put forward by the panel that groundwater is overlooked, for example, or that it requires specific advocacy, to problems of cooperation, data collection (monitoring) and data sharing. The panellists reiterated that Canada's complex policy and governance environment makes it necessary to be intentional about including groundwater in freshwater conversations. Economic arguments were recognized as potentially powerful and persuasive in helping drive home the importance of groundwater and freshwater generally.

The panel also noted that groundwater science has been historically underfunded by the federal government compared to surface water science: calling it out as an area of concern could help correct the imbalance. Finally, because groundwater is multi-jurisdictional and a multi-departmental responsibility, keeping it in the spotlight helps foster cross-government collaboration, which eliminates redundancies and makes departments more efficient.

The establishment of the Canada Water Agency was given as an example of real commitment to closing gaps in leadership and cooperation. In terms of data collection, the panel noted that policy is not the only tool required: technology and innovation are vitally important as well, for gathering more and better information and converting it into knowledge that can inform policy. Data sharing and data standards were agreed to be crucial, as their lack can pose very real barriers to effective resource management. Finally, it was repeated that a critical next step will be to design a national groundwater monitoring program or network of networks that is respectful of and rooted in regional needs and priorities.

DISCUSSION

The following discussion draws on arguments put forward by the panellists, contextualized by the 55-year history of federal freshwater policy. It focuses on five specific areas: 1) governance barriers, 2) funding, 3) capacity, 4) monitoring and 5) communication. Many of the issues and concerns raised can be traced back to the 1980s and documentation by the Inquiry on Federal Water Policy (Pearse et al. 1985) and numerous sub-



sequent publications. Despite their persistence, these issues and concerns are not static, shifting with changes in governments, evolving departmental priorities and the outcomes of policy reviews. The present discussion considers what has and what has not changed and identifies where targeted action is already occurring and could have the greatest effect.

Governance Barriers

Governance challenges related to water resources are rooted in the constitutional assignment of natural resources to the provinces and federal government and the fact that while the *Canada Water Act* emphasizes federal–provincial consultation and collaboration, it does not distinguish between surface water and groundwater. Overcoming these challenges will only be achieved through shared goals, collaboration and financial necessity. Generally, action on groundwater will require multi-stakeholder (e.g. Rivera et al. 2003; Schaefer and Bielak 2006) and multidisciplinary efforts (e.g. Rivera 2014). This challenge was recognized in the early 2000s by the Canadian Council of Ministers of the Environment which, in collaboration with National Water Research Institute, organized a series of water science–policy workshops (Schaefer and Bielak 2006).

The Senate Committee report, *Water in the West: Under Pressure* (Senate of Canada 2005), provides a powerful example of what political recognition of an issue and a sense of imperative can achieve. The report propelled groundwater progress on Canada's 30 key aquifers (Rivera 2014) for a 10-year period. By that example, progress *can* be made when politics, funding, science capacity and strong federal–provincial collaboration all coexist. Unfortunately, the achievements related to that work are not recorded in the *Canada Water Act* annual reports, even though efforts were aligned with the act and mirrored municipal water supply work completed previously by Environment Canada (1986). Momentum and funding for this national aquifer mapping effort ultimately waned, leaving much of Canada with unmapped groundwater bodies.

Up to 2024, the Canada Water Agency presented its focus as being on surface water while acknowledging groundwater is also within its scope (Goffin and Sullivan 2024). Since then, the agency has recognized the need for more dedicated attention to groundwater, and there are broader signs of groundwater gaining recognition, including in documents on freshwater science synthesis and the science agenda (Environment and Climate Change Canada 2024, 2026). The Canada Water Agency's increased emphasis on groundwater along with these wider acknowledgements demonstrates that formal, structural barriers can be overcome and that the culture of water science can change. What is now required is appropriate action by senior managers at all levels of government to bring about enhanced coordination and collaboration across the Government of Canada and with the provinces and territories. Furthermore, there needs to be involvement of groundwater experts in policy and management decision-making processes.

Funding

Funding for groundwater studies, and in fact for science in general, is captive to societal and political attention (e.g. Naylor

et al. 2017). Significantly and unfortunately, the Naylor et al. review of Canadian science excluded department-based government research. Nevertheless, the report recommended increased collaboration between government and academic researchers and between federal–provincial–territorial researchers. It also highlighted the need for improved inter-connected research within government to support policy.

These recommendations touch on issues that have been challenges for freshwater science since the inception of the *Canada Water Act* in 1970 and have been commented on over the life of the act (e.g. Interdepartmental Committee on Water 1990; Booth and Quinn 1995). Water resources support the larger environmental, economic and social issues that are often central to government policy; however, funding for proactive rather than reactive freshwater science, and more specifically groundwater science, is commonly overlooked. A now-classic example of this reactive versus proactive imbalance is the USA's Superfund program: created only after the 1970s discovery of sites like Love Canal, it has since required enormous investment in contaminant identification and remediation that proactive monitoring could have averted at a fraction of the cost.

In the late 1980s, declining attention to freshwater was attributed to the rise of the sustainable development perspective articulated in what is colloquially known as the 'Brundtland Report' (World Commission on Environment and Development 1987) and, subsequently, to the development of the Green Plan (see Government of Canada 1990; Booth and Quinn 1995). With Program Review in the mid-1990s, federal funding and capacity were both further diminished. For example, the National Hydrometric System was compelled to reduce its monitoring network by 30% (Environment Canada 2000).

Booth and Quinn (1995) have mapped out decadal funding variations and funding mechanisms during the first 25 years of the *Canada Water Act*. In the early 2000s, groundwater received minimal direct funding under the act even though the Geological Survey of Canada had been funded (Senate of Canada 2005) to work on 30 key aquifers, develop the Groundwater Information Network, and provide national synthesis (Rivera 2014; Natural Resources Canada 2019). This funding allocation had a period of 10 years, despite the monumental and iterative tasks required, and its implementation was made more complicated by shifting issues and growing population pressures.

These funding constraints and reductions over the years seem to have failed to recognize why water resource management—the principal pillar of the *Canada Water Act*—matters. Water is bound up inextricably with economic development, environmental protection and health. Testament to this, the first comprehensive review of groundwater in Canada was published as a Geological Survey of Canada Economic Geology report (Brown 1967). Additionally, in a report for the Prospector and Developers Association of Canada, Duke (2010) stated: "... *water resources may well become the most prominent resource issue of the 21st century.*" As noted previously, the pivotal role of freshwater in supporting all aspects of Canadian soci-

ety seems poorly appreciated in federal government departments. General funding as provided under the Freshwater Action Plan (Environment and Climate Change Canada 2025) is important; however, for the most part such funding is unavailable to federal government science departments. Consequently, while the government is supporting freshwater activity in major water basins, it is failing to provide commensurate support for government science infrastructure, staffing and science research. Both Environment and Climate Change Canada and Natural Resources Canada have been faced with declining core funding, resulting in staff attrition, challenges in maintaining research infrastructure and an inability to sustain thematic research initiatives.

The most substantial freshwater funding in decades was provided to the Canada Water Agency (2025b) in the 2023 federal budget for the Freshwater Action Plan (Department of Finance 2023). That plan, which is focused on managing, restoring and protecting freshwater in surface water basins, gives groundwater minimal profile. There is an opportunity to realign Canada's freshwater ecosystem priorities with the recently released Environment and Climate Change Canada Freshwater Science Agenda (Environment and Climate Change Canada 2026), which meaningfully includes groundwater science and research priorities, and to include nationally significant groundwater bodies as well.

Groundwater science needs stable, long-term funding for research and monitoring that supports the *Canada Water Act*. There is a need to build momentum by raising awareness among policymakers and the general public about groundwater—to develop expertise, fund science to characterize resources and establish a national groundwater monitoring network capable of providing long-term data, and to report on status and trends in line with international goals (e.g. the UN Sustainable Development Goals).

Capacity

The issue of scientific capacity in government is more complex than may first appear. Canada has recognized expertise in groundwater science but lacks the resources and infrastructure to engage in it comprehensively. While groundwater capacity has been improving in some provinces and territories and in certain municipalities, capacity in the federal government has diminished to historic lows unparalleled since the end of the Second World War. The decline in groundwater expertise at Environment and Climate Change Canada and Natural Resources Canada since the 1990s Program Review is provided in the IAH Canada (2024) brief to the Standing Committee on Environment and Sustainable Development. Loss of senior groundwater science expertise is particularly critical, as even capable researchers and investigators will struggle to make progress without strong leadership driving action. The challenge of leadership loss has been illustrated by de Loë and Patterson (2017), who found that water-focused science staff have minimal impact when their efforts are held up against the larger and more powerful socio-economic perspectives exterior to the water community.

The federal workforce adjustment exercise underway today combined with emerging governmental priorities and budgetary constraints does not make the prospect of addressing the capacity gap particularly promising. An analysis of data from Statistics Canada and the Council of Chairs of Canadian Earth Science Departments concluded the number of geoscientists in the labour market declined by 11% between 2011 and 2021, with university enrolment in geoscience programs declining in many countries including Canada (Lebel et al. 2025).

This decline is part of a broader North American trend and may be attributable to the perception of job instability in the natural resources sector and to misconceptions that the geosciences are limited to oil, gas and mining (Lebel et al. 2025). The lack of visibility of Earth sciences in high school curricula may also be a factor, along with competition from programs in environmental science, climate science and engineering. All of these are linked, directly or indirectly, to a lack of interest in and a lack of understanding of the Earth sciences, which include groundwater, implying a broad public failure to recognize its importance. Also related to education, water policy programs should incorporate more extensive groundwater subject matter, and hydrogeology programs could integrate water policy components to help create a pipeline of 'groundwater literate' future leaders.

The takeaway is that building leaders' knowledge and awareness of groundwater and their ability to make water-minded policy is as important as building raw science capacity. This can be achieved by integrating scientists into policy-making and other decision-making processes, with increased representation of groundwater expertise in freshwater discussions. Particularly when budgets are tight, it may be advisable for government to engage with hydrogeology experts through professional societies such as IAH Canada, including its Committee for Groundwater Advocacy, which conducts groundwater policy research. Successful development of groundwater within freshwater public science funding and much-needed groundwater science likely requires groundwater issues to be linked explicitly to political, educational and economic priorities, i.e. by reframing the perception of groundwater so that it is seen as being integral to water security and economic growth.

Monitoring

Canada is the only G7 and circumpolar country without a national monitoring program for groundwater (Ryan et al. 2026). This lack of infrastructure is problematic as it renders Canada unable to monitor interprovincial (regional) groundwater changes. It also constrains groundwater science and handicaps federal reporting for international commitments. As noted earlier in this article, Canada has a world-class national surface water monitoring network of approximately 3 000 stations (Environment and Climate Change Canada 2025), yet only nine groundwater sites in the National Hydrometric Network. Groundwater monitoring is nearly completely implemented on the provincial, watershed and local scale. For the provinces and territories, water level monitoring networks are

maintained, documented and delivered online to various levels of completeness for nine provinces and one territory.

While noteworthy academic groundwater networks exist at research stations and watersheds (e.g. Larocque and Thériault 2026), these are designed, implemented and maintained to achieve specific research objectives, and not to record general observations over time. Academic networks also commonly have short funding cycles, whereas multidecadal funding is needed to sustain and maintain a national groundwater monitoring network.

The challenge of groundwater monitoring was raised by the Interdepartmental Committee on Water over 30 years ago (1990). The need for data collection under an open access model was further raised by the National Round Table on the Environment and the Economy (2012) through its National Water Forum. There has been little subsequent progress, even though there is no legislated or policy directive that the National Hydrometric Program should be focused on surface water monitoring (e.g. Environment and Climate Change Canada 2023). The panel discussion highlighted that the importance of monitoring goes beyond resource management, and in fact comprises a form of critical infrastructure for freshwater science, especially in the context of climate change. A review of briefs submitted to the freshwater committee of the House of Commons Standing Committee on Environment and Sustainable Development indicated concern regarding groundwater monitoring and a need for national (federal) involvement for groundwater quantity, quality, watershed and surface water interaction issues.

An interim solution has been to develop a data system that supports a ‘network of networks’ approach by the Groundwater Information Network (e.g. Brodaric et al. 2016), which delivers water well and monitoring well data from multiple provinces using international standards. However, the Groundwater Information Network approach does not touch on the purposeful design of a national network or standards for water quality, well placement, well design, measurement devices, and more, that are found in the National Hydrometric Service surface water system (Environment and Climate Change Canada 2023) and US National Groundwater Monitoring Network (Subcommittee on Ground Water 2013).

Developing a true national network likely requires integration of groundwater representation at the National Administrator’s Table and the National Hydrometric Program, which are currently attended by provincial and territorial representatives of departments responsible for both surface water and groundwater. It also requires establishing a funded mandate to design and implement a network, a point highlighted in 1990 (Interdepartmental Committee on Water 1990).

Jurisdictional differences may be another challenge to overcome. Panellists and audience participants acknowledged that developing a national network could be complicated by conflicting perspectives, objectives and provincial and territorial capacity. Nevertheless, all agreed there are strong grounds for arguing in favour of a national monitoring approach, with well documented results and cost benefits of monitoring the water

cycle overall and groundwater specifically (e.g. Kim and Kim 2019).

Communication

A relatively small proportion of Canadians understand what groundwater is or appreciate its importance. Beyond bureaucrats and politicians, the general public’s ‘groundwater literacy’ needs bolstering. Those who do recognize why and how groundwater matters must raise awareness that freshwater includes both groundwater and surface water. Communications need to reflect this fact. As technical jargon can be a barrier when trying to reach non-specialist audiences, plain language communication efforts will be important to any educational push.

The unspecific language of the *Canada Water Act* may be adequate for individuals with deep knowledge of the integrated nature of the water cycle but likely does not register with non-specialists. This is a major challenge for groundwater gaining recognition and appropriate attention. As de Loë and Patterson (2017) highlighted, it may be difficult to change the behaviour of non-specialists if they feel their primary interests and objectives might be compromised by doing so. For example, failure to understand groundwater’s contribution to base-flow and thermal refugia for spawning fish (or in delivering contaminants to critical aquatic habitat regions) may not be obvious to many who want to divert groundwater resources for higher value economic returns.

Increased communication, information and data sharing will therefore be key to the success of any multidisciplinary approach, and efforts are underway. The panel discussion itself marked an important step in this direction, as did the earlier publication of *Canada’s Groundwater Resources* (Rivera 2014), highlighted by Harford (2015) as an invaluable compendium of groundwater science aimed at a public audience. An excellent Canadian not-for-profit academic example is *The Groundwater Project* (Cherry 2020), while more public-oriented approaches are offered by AquaAction (<https://aquaaction.org>) and the Canadian Geographic Society (<https://canadaoceanmap.ca>).

Several provinces have launched annual groundwater geoscience workshops to communicate government work to the public and technical users. In collaboration with the Geological Survey of Canada and other organizations, the Ontario Geological Survey has been running an annual workshop and publishing abstracts from its proceedings for the past decade (e.g. Burt et al. 2026). An annual hybrid workshop was initiated in 2022 in British Columbia. The province is also publishing a quarterly groundwater-focused newsletter (Government of British Columbia 2026).

Both IAH Canada and the Canadian Water Resources Association host podcasts and local and regional workshops for groundwater professionals. The “Réseau québécois sur les eaux souterraines” (<https://www.rqes.ca>) makes groundwater knowledge, research results, educational tools and publications publicly accessible to support informed decision-making about Québec’s aquifers.

As more participants get involved and work together, and as more data and knowledge are shared, efforts can be pooled to build up better understanding of groundwater resources. Any such communication and data-sharing must be formalized and must also be encouraged and financially supported by governments: it cannot be expected to happen on its own.

CONCLUSIONS

Bringing scientists, policymakers, and decision makers together to advance practical aims remains a challenge given their different operational timeframes and knowledge and, in many cases, cultures, values and ethics (Schaefer and Bielak 2006). Yet it seems undeniable that researchers and policymakers could benefit from an increased understanding of each other's disciplines, objectives and constraints. The 2025 groundwater panel helped breach this divide by giving a predominantly science-oriented audience the opportunity to hear from individuals representing policy and senior management perspectives. The panel also stood out for its collective involvement of Natural Resources Canada, the Canada Water Agency and provincial, watershed and international NGO representatives, potentially signalling a new era in communication and collaboration on groundwater.

Four specific outcomes would help address many of the concerns raised during the panel discussion:

1. Explicit inclusion of groundwater in any renewed federal freshwater policy framework.
2. Stable, multi-year funding for groundwater science and monitoring structured to build institutional capacity and long-term datasets rather than fund discrete projects—coordinated through the Canada Water Agency and administered in partnership with the Geological Survey of Canada, Environment and Climate Change Canada, provincial geoscience and environmental bodies, Indigenous communities and NGOs.
3. Development of a national monitoring network co-led by the Geological Survey of Canada and Environment and Climate Change Canada, with provincial and territorial partners and integrated into the existing National Hydro-metric Program for surface water.
4. Formalization of groundwater expertise within federal freshwater decision-making processes, including representation at the National Administrators Table and explicit groundwater workstreams within the Canada Water Agency's thematic priorities.

In many ways, these panel outcomes parallel IAH Canada's recommendations to the Standing Committee on Environment and Sustainable Development and the findings of that committee's final report (House of Commons of Canada 2026). Specifically, the committee called on the Canada Water Agency to dedicate attention and resources to the role of groundwater in water security and climate adaptation, for groundwater to be built into a renewed *Canada Water Act*, and for freshwater monitoring that includes groundwater, empha-

sizing the need for analysis related to human health impacts (House of Commons of Canada 2026).

The panel at the national meeting in 2025 of the Geological Association of Canada – Mineralogical Association of Canada meeting in Ottawa jointly coordinated with IAH Canada demonstrated that the scientific community, federal and territorial governments and conservation practitioners share a common sense of direction. The challenge is to sustain that shared sense and find the broader support to convert it into policy, monitoring infrastructure and sustained investment. Canada's groundwater, and the surface water, ecosystems and communities and economic sectors that depend on it, cannot afford another 35-year interval between recognition and action.

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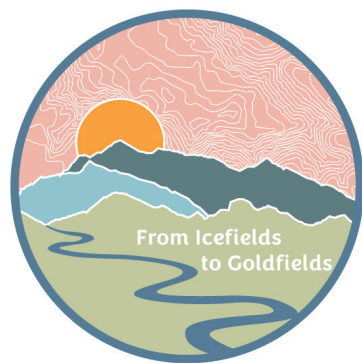
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GAC-MAC 2027

June 13-16

Whitehorse, Yukon

SECOND CALL FOR PROPOSALS!

The local organizing committee is still accepting proposals for symposia, special sessions, field trips, short courses, and workshops for next year's GAC-MAC Meeting in Whitehorse, Yukon. Want to be part of organizing some fun? Don't delay and get your proposal submitted today!

Proposal submissions are off to a great start! Sessions on a variety of critical metal, gold, and gem deposit types, stratigraphy, Quaternary volcanism, and stratigraphic palaeontology have been submitted in addition to several field trips highlighting northern Cordilleran geology.



Fall foliage in the St. Elias Mountains, Yukon

Select field trips include:

- Tombstone Gold Belt: Classic and new reduced intrusion-related gold deposits
- Volcanogenic massive sulphide and orogenic gold deposits in Southeast Alaska
- Quaternary subglacial and explosive volcanism in the Canadian Cascades arc, British Columbia

The extended deadline to submit a proposal is November 1, 2026. To submit a proposal, fill in the form at <https://forms.gle/3NXonTLuvpFf5me88> or email gacmac2027@gmail.com with a name and short description of your session or event, along with contact information for convenors/organizers.

A preliminary program with session and event information will be circulated on the conference website in December 2026. Abstract submission will be open from December 2026 to March 2027. Conference registration will open in February 2027.